

AI Service Detailed Explanation

Ai-VendorHub Marketplace | Updated Current Capabilities

Updated after latest AI implementation: marketplace-limited scope, all routes mounted, centralized prompts, proper fallback messages.

1. Overview

Tumhara AI Service ab Ai-VendorHub marketplace ke liye limited AI assistant hai. Ye general-purpose chatbot nahi hai. Ye sirf ecommerce marketplace tasks handle karega: product search, recommendations, similar products, comparison, budget shopping, mood shopping, seller descriptions, category/tags, and review summaries.

Agar user off-topic question puche, jaise politics/weather/general knowledge, AI answer nahi karega. Instead proper message dega ki AI sirf Ai-VendorHub marketplace tasks mein help kar sakta hai. Agar request unclear ho, AI clarifying message dega.

2. Active Endpoints

Feature	Endpoint	Auth	Current Status
AI Product Search	POST /ai/search-intent	Required	Mounted and marketplace scoped
Conversational Shopping	POST /ai/chat	Required	Mounted with domain guard
Similar Products	GET /ai/similar/:productId	Required	Mounted
Product Comparison	POST /ai/compare	Required	Mounted
Smart Budget Shopping	POST /ai/smart-budget	Required	Mounted
Mood Shopping	POST /ai/mood-shopping	Required	Mounted
Description Generator	POST /ai/generate-description	Public currently	Mounted with safe fallback
Category & Tags	POST /ai/suggest-category-tags	Public currently	Mounted with safe fallback
Review Summary	POST /ai/review-summary/:productId	Required	Mounted with fallback
Metrics	GET /ai/metrics	Public currently	Mounted
Feature Flags	GET/POST /ai/feature-flags	Public currently	Mounted
AI Scope	GET /ai/scope	Public currently	Mounted

3. Marketplace Scope Control

Situation	AI Behavior
User asks shopping/product query	AI processes request and calls Product Service where needed.
User asks seller listing task	AI generates description/category/tags using marketplace-safe prompt.
User asks unclear query	AI returns clarification message asking for product/category/budget/use-case.
User asks off-topic query	AI refuses politely and redirects to marketplace tasks.
Product Service unavailable	AI returns Product Service unavailable message instead of crashing.
Gemini unavailable	Fallback/local parser/basic generation is used where possible.

4. Centralized Prompt System

Prompts ab services mein scattered hardcoded form mein nahi hain. Prompt logic centralized hai in ``Al/src/services/prompt.service.js``. Isse future mein prompt update easy hoga aur env variables se override bhi possible hai.

Prompt Area	Purpose
marketplaceClassifier	Request marketplace scope mein hai ya off-topic/unclear hai, classify karta hai.
searchIntent	Natural language query ko product filters mein convert karta hai.
searchSummary	Found products ke basis par buyer-friendly summary banata hai.
description	Seller product description, bullets, tags, SEO keywords generate karta hai.
categoryTags	Product category, subcategory, tags, confidence suggest karta hai.
reviewSummary	Reviews se pros, cons, sentiment, recommendation score banata hai.
ecommerceAgentSystem	Socket/agent assistant ko marketplace-only rules deta hai.

5. AI Product Search

Buyer simple language mein search karega. AI query ko filters mein convert karega and Product Service se real products fetch karega.

```
POST /ai/search-intent
Authorization: Bearer <token>

{
  "query": "show me gaming headphones under 3000"
}
```

Current Ability	Usefulness
Keywords/category/budget/sort intent detect karta hai.	User ko filters manually select nahi karne padte.
Product Service env URL use karta hai.	Deployment mein localhost hard dependency reduce hoti hai.
Off-topic/unclear detection added.	AI project scope ke bahar answer nahi karega.
No products found case handled.	User ko helpful retry suggestion milta hai.

6. Conversational Shopping Assistant

Endpoint ``/ai/chat`` now active hai. Ye chat-style shopping assistant hai jo product search/recommendation conversation handle karta hai. Domain guard laga hua hai, so ye sirf marketplace queries handle karega.

```
POST /ai/chat
Authorization: Bearer <token>

{
  "message": "I need headphones for gym under 3000",
  "sessionId": "user-session-1"
}
```

7. Recommendation Features

Feature	Endpoint	What It Does
Similar Products	GET /ai/similar/:productId	Current product ke title/category/description se related products suggest karta hai.
Product Comparison	POST /ai/compare	2+ products ko price, stock, category, description ke basis par compare karta hai.
Smart Budget Shopping	POST /ai/smart-budget	Budget and purpose ke according product bundle suggest karta hai.
Mood Shopping	POST /ai/mood-shopping	Mood/vibe jaise gaming, cozy, study, minimal ko product keywords mein map karta hai.

8. Seller AI Features

Feature	Endpoint	How Seller Uses It
Description Generator	POST /ai/generate-description	Seller title/basic details dekar professional description, bullets, tags, SEO keywords generate kar sakta hai.
Category & Tags	POST /ai/suggest-category-tags	Product title/description se category, subcategory, tags, confidence score milta hai.

In dono features mein fallback logic bhi add hai. Agar Gemini unavailable ho, AI basic safe marketplace content generate karega instead of failing.

9. Review Summary

Review summary buyer decision making ke liye hai. Product reviews ko pros, cons, sentiment, summary, and recommendation score mein convert karta hai. Agar reviews nahi hain, clear message return karta hai.

POST /ai/review-summary/:productId
Authorization: Bearer <token>

10. Metrics and Feature Flags

Endpoint	Purpose
GET /ai/metrics	LLM calls, success/failure, latency, fallbacks, feature flags return karta hai.
GET /ai/feature-flags	Current LLM feature flags show karta hai.
POST /ai/feature-flags	Runtime feature flags update karne ka endpoint.
GET /ai/scope	AI ka allowed marketplace scope return karta hai.

11. Reliability Improvements

Improvement	Impact
Domain Guard	AI ko project marketplace tak limited rakhta hai.
Centralized Prompts	Prompt maintenance easy and clean.
JSON Extract Utility	LLM response parsing safer hua.
Circuit Breaker	Repeated Gemini failures se service protected.
Retry With Backoff	Temporary LLM issues handle.
Fallback Responses	AI unable/unclear/product-service-down cases mein clean response.
JWT Auth Middleware	Buyer-facing AI endpoints protected.

12. Example Proper Messages

Case	Message Style
Unclear request	I could not clearly understand what product you want. Please mention product type, category, budget, or use-case.
Off-topic request	I can help only with Ai-VendorHub marketplace tasks like product search, recommendations, comparisons, product descriptions, category/tag suggestions, and review summaries.
Product Service down	I understood your shopping request, but Product Service is not reachable right now. Please try again after starting Product Service.
No reviews	No reviews found for this product, so AI summary is not available yet.

13. Final Status

AI Service ab complete project-level marketplace assistant hai. Saare advanced AI controllers routes par mounted hain. AI ab general chatbot nahi hai; ye Ai-VendorHub marketplace/product/seller tasks tak limited hai. Prompt management centralized hai, fallback handling better hai, and authenticated buyer AI endpoints protected hain.

Important setup note: AI `.env` mein `JWT\_SECRET` Auth service ke JWT secret se same hona chahiye, warna protected AI routes token reject karenge. `GOOGLE\_API\_KEY` missing hone par LLM feature flags off ho jayenge, but fallback logic where possible still response dega.